Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01]

Floors, ceilings, fractional parts, and absolute values replace one formula by several exact pieces. Write the domain and range of each piece with its open or closed endpoints, then count images or preimages. Separate integer-count arguments from area arguments, where endpoints have zero measure.

取整、上取整、小数部分与绝对值会把一个公式拆成若干精确分段。须写清每段定义域、值域及开闭端点，再计像或原像。整数计数与面积论证应区别处理，后者端点测度为零。

Invert a floor into an exact half-open interval

将取整反解为精确半开区间

Procedure

1. Replace $\lfloor u \rfloor = k$ by $k \le u < k+1$, preserving both endpoint signs.

2. Invert a monotone expression only after fixing its sign and domain.

3. Count integers using floor and ceiling bounds, or count branches whose ranges contain the target.

4. Prove the converse: every counted value or input really lies in its claimed interval.

通用步骤

1. 把 $\lfloor u \rfloor = k$ 换成 $k \le u < k+1$，保留两端开闭。

2. 固定符号与定义域后才反解单调表达式。

3. 用上下取整计算整数，或计算值域包含目标的分支。

4. 证明逆向成立：每个计入的值或输入确实落在对应区间。

> Example. Find all integers x with $\lfloor (2x+1)/5 \rfloor = -2$.

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绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01] (continued / 续)

> 例题：求满足 $\lfloor (2x+1)/5 \rfloor = -2$ 的全部整数 x。

Worked solution

1. The floor condition is $-2 \le (2x+1)/5 < -1$.

2. It becomes $-11 \le 2x < -6$, or $-11/2 \le x < -3$.

3. The integer solutions are $-5, -4$. In particular, the upper endpoint -3 is excluded.

例题解析

1. 取整条件为 $-2 \le (2x+1)/5 < -1$。

2. 整理成 $-11 \le 2x < -6$，即 $-11/2 \le x < -3$。

3. 整数解为负五、负四，特别是上端点负三不包含。

Floor means rounding downward, including for negative inputs.

下取整始终向下，负数也一样。

For $A > 0, n > 0$ and positive k, $\lfloor A/n \rfloor = k$ means $A/(k+1) < n \le A/k$. A half-open interval of length at least one always contains an integer. For $x\lfloor x \rfloor$ with $a = \lfloor x \rfloor \ge 1$, the exact range is $[a^2, a^2 + a)$ and each integer in it has inverse $x = N/a$.

For an integer $j \ge 0$, $f(x) = x\{x\}$ on $[j, j+1)$, writing $x = j + t$ gives $f = (j+t)t$, strictly increasing onto $[0, j+1)$. Iterated preimages can therefore be counted by weakly ordered branch indices. Integer endpoints map to zero and must be checked separately.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01] (continued / 续)

在 $A>0, n>0, k>0$ 时，$\lfloor A/n \rfloor = k$ 等价于 $A/(k+1) < n \le A/k$。长度至少一的半开区间必含整数。对 $x\lfloor x \rfloor$，若 $a = \lfloor x \rfloor \ge 1$，精确值域为 $[a^2, a^2+a)$，其中每个整数 N 的逆像为 $x = N/a$。

对整数 $j \ge 0$，在 $[j, j+1)$ 上，令 $x = j+t$，则 $x\{x\} = (j+t)t$ 严格递增且值域为 $[0, j+1)$，因此迭代原像可由弱递增分支下标计数。整数端点映到零，须另行检验。

Practice / 配套练习: AIME-2002-II-Q08 (WB p. 162), AIME-2007-II-Q07 (WB p. 164), AIME-2009-I-Q06 (WB p. 166), AIME-2009-II-Q11 (WB p. 174), AIME-2012-II-Q10 (WB p. 168), AIME-2015-II-Q13 (WB p. 203), AIME-2020-II-Q14 (WB p. 178), AIME-2022-II-Q08 (WB p. 504)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02]

Group rounding by thresholds and periodic residue classes

按阈值与周期剩余类归组四舍五入

Procedure

1. Write the exact interval producing each rounded value, including any tie convention.

2. If changing the input by a fixed period changes intermediate rounded values by integers, prove a residue-class success pattern.

3. For root rounding or square-free blocks, compare consecutive square gaps with the block width.

4. Count complete blocks and the truncated final block separately.

通用步骤

1. 写出产生每个取整值的精确区间，并说明等距时的规则。

2. 若输入加固定周期使中间取整值增加整数，先证明成功模式按剩余类周期重复。

3. 平方根取整或无平方数区块可比较相邻平方差与区块宽度。

4. 分开计算完整区块与最后截断块。

> Example. For integer n, let $R(n)$ be the nearest integer to $n/3$, and then round $3R(n)$ to an integer. Among $0 \le n \le 20$, which inputs return unchanged?

> 例题：对整数 n，将 $n/3$ 四舍五入为整数 $R(n)$，再取整数 $3R(n)$。零至二十中哪些输入能回到原数?

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02] (continued / 续)

Worked solution

1. Write $n = 3q + r$, with $r = 0, 1, 2$. The rounded values are $q, q, q + 1$ respectively, with no ties.

2. The returned integers are $3q, 3q, 3q + 3$, so equality holds exactly when $r = 0$.

3. The successful inputs are $0, 3, 6, 9, 12, 15, 18$, seven in total. The residue calculation proves the pattern globally.

例题解析

1. 写 $n = 3q + r$，$r = 0, 1, 2$ 时取整分别为 $q, q, q + 1$，无等距情形。

2. 返回整数分别为 $3q, 3q, 3q + 3$，仅余数零时等于原数。

3. 成功输入为零、三、六、九、十二、十五、十八，共七个；剩余类计算证明了整体周期。

A periodic rounding rule still needs an endpoint count in a finite input interval.

周期取整规则在有限输入范围内仍须处理端点截段。

For positive integers n, k, the nearest integer to $\sqrt{n}$ is k precisely when $(k - 1/2)^2 < n < (k + 1/2)^2$. This becomes $k^2 - k + 1 \le n \le k^2 + k$, with $2k$ inputs. No square-root tie occurs because the squared half-integer endpoints are not integers. The case $n = 0$ gives rounded value zero separately.

A tuple of nondecreasing integer floors stays constant on a consecutive block. To be uniquely identified by it, an integer must have a tuple jump immediately before and after it. This converts singleton floor blocks into divisibility of two consecutive integers.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02] (continued / 续)

对正整数 n, k，$\sqrt{n}$ 最近整数为 k 恰要求 $(k-1/2)^2 < n < (k+1/2)^2$，即 $k^2 - k + 1 \le n \le k^2 + k$，共有 $2k$ 个输入。半整数端点的平方非整数，因此无取舍平局；$n = 0$ 时另有取整值零。

若干非降整数取整组成的向量在连续整数区块上恒定。要由该向量唯一确定某整数，它前后必须都发生向量跳跃，可转为相邻整数的整除条件。

Practice / 配套练习: AIME-2007-I-Q05 (WB p. 161), AIME-2007-I-Q11 (WB p. 172), AIME-2008-I-Q07 (WB p. 165), AIME-2013-II-Q06 (WB p. 548), AIME-2022-II-Q08 (WB p. 504)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 3 / 方法 3 [AL06-M03]

Read logarithmic floors as geometric bands or indicators

把对数取整读成等比区间或指示量

Procedure

1. Translate logarithm thresholds into powers of the base.

2. A floor residue condition selects disjoint bands; sum their lengths with a geometric series.

3. Use $\lceil u \rceil - \lfloor u \rfloor$ as zero at integers and one elsewhere.

4. For a monotone floor sum, certify an extremum by evaluating the last legal input and the next one.

通用步骤

1. 将对数阈值转成底数的幂。

2. 取整剩余类条件选择不交区间带，用几何级数求其长度。

3. 把 $\lceil u \rceil - \lfloor u \rfloor$ 视为整数处零、其他处一的指示量。

4. 单调取整和的最值可由最后合法输入与紧接的下一个输入核验。

> Example. For uniformly chosen $x \in (0, 1]$, find the probability that $\lfloor \log_3(1/x) \rfloor \equiv 1 \pmod 3$.

> 例题：在 $(0, 1]$ 均匀选 x，求 $\lfloor \log_3(1/x) \rfloor$ 模三余一的概率。

Worked solution

1. The permitted floors are $3j + 1$, giving $3^{-3j-2} < x \leq 3^{-3j-1}$ for $j \geq 0$.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 3 / 方法 3 [AL06-M03] (continued / 续)

2. Each interval length is $(1/3 - 1/9)3^{-3j} = (2/9)27^{-j}$.

3. The total length, hence probability, is $(2/9)/(1 - 1/27) = 3/13$.

例题解析

1. 允许的取整值为 $3j + 1$，对应 $3^{-3j-2} < x \leq 3^{-3j-1}$。

2. 每段长度为 $(2/9)27^{-j}$。

3. 总长度即概率，为 $(2/9)/(1 - 1/27) = 3/13$。

For an integer argument, an integral logarithm may force a special power family; prove which powers are themselves integers.

整数自变量的对数为整数时可能只剩特殊幂族，须证明哪些幂本身为整数。

Practice / 配套练习: AIME-2004-I-Q12 (WB p. 171), AIME-2007-I-Q07 (WB p. 163), AIME-2010-I-Q14 (WB p. 175)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 4 / 方法 4 [AL06-M04]

Separate a floor sum into its smooth sum and exact residues

将取整和拆成未取整和与精确余数

Procedure

1. Use $\lfloor a/m \rfloor = (a-r)/m$ with the least residue $0 \le r < m$.

2. First bound the total floor error to isolate any unknown integer parameter.

3. Then compute the exact residue correction by full periods and a final partial period.

4. For piecewise-linear area sums, track the fractional part of each unit-strip contribution in the same way.

通用步骤

1. 用 $\lfloor a/m \rfloor = (a-r)/m$，其中最小非负余数为 r。

2. 先估计取整误差总量，隔离可能的未知整数参数。

3. 再按完整周期与尾部截段计算精确余数修正。

4. 分段线性面积和也可同样追踪每个单位条带贡献的小数部分。

> Example. Compute $\sum_{n=1}^{10} \lfloor (3n+2)/4 \rfloor$.

> 例题：计算 $\sum_{n=1}^{10} \lfloor (3n+2)/4 \rfloor$。

Worked solution

1. The unfloored sum is $(3 \cdot 55 + 20)/4 = 185/4$.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 4 / 方法 4 [AL06-M04] (continued / 续)

2. The residues of $3n+2$ modulo four repeat as $1, 0, 3, 2$. Two full blocks contribute twelve, and the last two terms add one.

3. Subtract the residue total divided by four: $(185-13)/4 = 43$.

例题解析

1. 未取整和为 $(3\cdot 55+20)/4 = 185/4$。

2. $3n+2$ 模四的余数循环为一、零、三、二，两整块贡献十二，最后两项再加一。

3. 扣除余数总量除四，得到 $(185-13)/4 = 43$。

A uniform error bound can prove uniqueness of a parameter before a large sum is evaluated exactly.

统一误差界可以先证明参数唯一，再求大和的精确值。

If N floors each subtract a fractional part in $[0,1)$, their total error lies in $(-N,0]$. If each unit increase of an integer parameter changes the smooth sum in the same direction by more than the permitted window plus this error width, only one parameter can work. For areas under $x\lfloor\sqrt{x}\rfloor$, each integer strip contributes $k(2j+1)/2$; its half-integer parity depends only on whether k is odd.

N 个下取整各减去 $[0,1)$ 内的小数部分，误差总量在 $(-N,0]$。若整数参数每增加一都使未取整和向同一方向变化，且变化量超过允许窗口加误差宽度，就至多一个参数可行。对 $x\lfloor\sqrt{x}\rfloor$ 下的面积，每个整数条带贡献 $k(2j+1)/2$，其半整数奇偶仅取决于 k 是否为奇数。

Practice / 配套练习: AIME-2015-I-Q14 (WB p. 177), AIME-2023-I-Q10 (WB p. 170)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05]

Resolve signs and count absolute-value levels

拆分符号并计算绝对值水平线根数

Procedure

1. List all breakpoints where an absolute-value input changes sign.

2. Replace the expression by its exact formula on each interval, retaining its interval restrictions.

3. Construct a level-count table at regular levels, turning levels, zero, and any flat interval.

4. For nested absolute values, translate the outer equation into distinct inner levels and sum their valid root counts.

通用步骤

1. 列出绝对值内部变号的全部分界点。

2. 在各区间用精确公式代替，并保留该区间限制。

3. 对普通高度、转折高度、零高度及平坦段建立根数表。

4. 嵌套绝对值方程先转成不同内部高度，再相加其有效根数。

> Example. Solve $|x-1|+|x+2|=5$, and explain what happens if the right side is three.

> 例题：解 $|x-1|+|x+2|=5$，并说明右端改为三时有何变化。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05] (continued / 续)

Worked solution

1. The breakpoints are $-2, 1$. The expression is $-2x - 1$ to the left, three in the middle, and $2x + 1$ to the right.

2. At level five the left and right roots are $-3, 2$, both in their legal intervals; the middle gives none.

3. At level three every $x \in [-2, 1]$ works, so the solution set is an interval rather than a finite root count.

例题解析

1. 分界为负二、一，表达式左段为 $-2x - 1$，中间恒三，右段为 $2x + 1$。

2. 高度五时左右根为负三、二，均在合法段内，中段无解。

3. 高度三时整个 $[-2, 1]$ 都是解，不能按有限根数处理。

Squaring or dropping interval restrictions can turn an illegal branch root into a false solution.

平方或丢掉区间限制，会把无效分支根变成伪解。

An absolute-value preimage $|x - a| = z$ exists only for $z \geq 0$: it gives $a \pm z$, with the two values coinciding at zero. In a two-variable factor-sign region, consider all sign combinations and impose the original plane or ordering constraints before declaring a region bounded.

For an ordered zero-sum tuple with a fixed sum of absolute values, split into total positive and negative mass. An ordered coordinate of rank k cannot carry more than the corresponding mass divided by the number of coordinates forced to share its sign and magnitude bound. The equality tuple must be checked separately.

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绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05] (continued / 续)

绝对值原像 $|x-a|=z$ 只在 $z \geq 0$ 时存在，产生 $a \pm z$，零时两值重合。两个变量的乘积符号区域须检查全部符号组合，并施加原平面或次序条件后，才能判断区域有界。

总和为零且绝对值和固定的有序元组，可拆成正质量与负质量。某排名坐标的大小不超过相应质量除以被迫具有同号同量级界的项数；等号元组仍须独立检验。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206), AIME-2010-I-Q11 (WB p. 458), AIME-2020-II-Q08 (WB p. 209), AIME-2021-I-Q08 (WB p. 169), AIME-2022-II-Q06 (WB p. 184), AIME-2025-I-Q12 (WB p. 413)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 6 / 方法 6 [AL06-M06]

Treat floor graphs as cells and strip areas

把取整图形视为单元格与条带面积

Procedure

1. Fix every floor value and translate it into a half-open cell or strip.

2. Within that piece, solve the remaining smooth condition and intersect its range with the cell.

3. For area or probability, sum disjoint piece measures; isolated boundary points have zero measure.

4. For a closed enclosing object, pass to the closure and prove its extremal boundary points are included in the required bound.

通用步骤

1. 固定各个取整值，转成半开单元格或条带。

2. 在分段内解余下光滑条件，并与该格取交。

3. 面积概率按不交分段求和，孤立边界点测度为零。

4. 闭包围对象须包含原集合闭包，并证明极端边界点给出必要界。

> Example. Inside $[0,2)^2$, let R satisfy $\lfloor x \rfloor + \lfloor y \rfloor = 1$. Find its area and the radius of its smallest closed enclosing disk.

> 例题：在 $[0,2)^2$ 中令 R 满足 $\lfloor x \rfloor + \lfloor y \rfloor = 1$，求面积与最小闭包围圆盘半径。

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绝对值、取整与分段图像

Method 6 / 方法 6 [AL06-M06] (continued / 续)

Worked solution

1. The two cells have floor pairs $(0,1),(1,0)$, each of area one, so the area is two.

2. Any closed enclosing disk contains their closures, including opposite points $(0,2)$ and $(2,0)$, whose distance is $2\sqrt{2}$. Thus its radius is at least $\sqrt{2}$.

3. The disk centered at $(1,1)$ of radius $\sqrt{2}$ contains both closed squares, attaining the lower bound.

例题解析

1. 取整对为 $(0,1),(1,0)$ 的两个单元格各面积一，总面积二。

2. 闭包围圆盘须含两格闭包，包括距离 $2\sqrt{2}$ 的对点 $(0,2),(2,0)$，故半径至少 $\sqrt{2}$。

3. 以 $(1,1)$ 为心、半径 $\sqrt{2}$ 的圆盘包含两闭正方形，达到下界。

A boundary excluded from a half-open set can still force the radius of every closed containing disk.

半开集排除的边界仍可能限制所有包含它的闭圆盘半径。

Practice / 配套练习: AIME-2010-I-Q08 (WB p. 167), AIME-2011-II-Q15 (WB p. 176), AIME-2015-I-Q14 (WB p. 177)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 7 / 方法 7 [AL06-M07]

Filter periodic-piece roots before applying Vieta

先筛周期分段根再使用韦达

Procedure

1. Bound the possible function values, then use the other equation to bound the number of relevant periods.

2. Write one polynomial equation for each rising or falling piece.

3. Keep only roots satisfying that piece domain; count shared endpoints once.

4. Apply Vieta only to a complete retained root pair, and add any single surviving endpoint root separately.

通用步骤

1. 先限制函数值，再用另一方程限制相关周期数量。

2. 对每个上升或下降分段写一个多项式方程。

3. 仅保留符合该分段定义域的根，共用端点只计一次。

4. 只有完整保留的一对根才能直接用韦达求和，单个幸存根须另外相加。

> Example. Let a period-two triangular wave equal x on $[0,1]$ and $2-x$ on $[1,2]$. Find its intersections with $y=x/3$ for $x \geq 0$.

> 例题：周期二的三角波在 $[0,1]$ 为 x、在 $[1,2]$ 为 $2-x$，求其与 $y=x/3$ 在非负横轴上的交点。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 7 / 方法 7 [AL06-M07] (continued / 续)

Worked solution

1. The wave lies in $[0,1]$, so $0 \le x \le 3$. Only pieces $[0,1], [1,2], [2,3]$ matter.

2. The piece equations are $x = x/3$, $2 - x = x/3$, and $x - 2 = x/3$, giving $x = 0, 3/2, 3$ respectively.

3. All lie in their pieces, so the intersections are $(0,0), (3/2, 1/2), (3,1)$. The endpoint at three is counted once.

例题解析

1. 波值在零与一之间，所以横坐标只须考虑零至三，共三段。

2. 三段方程为 $x = x/3$、$2 - x = x/3$、$x - 2 = x/3$，分别得到 $x = 0, 3/2, 3$。

3. 都位于各自合法段，所以交点为 $(0,0), (3/2, 1/2), (3,1)$，端点三只计一次。

A polynomial may have two roots while its original piece retains only one.

多项式可能有两根，但原分段可能只保留一根。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206), AIME-2025-I-Q11 (WB p. 180)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08]

Count events in a finite window under stepwise constraints

在阶梯限制下计算有限窗口事件数

Procedure

1. Replace a ceiling-dependent rule by intervals where its integer value is fixed.

2. Optimize the rate or capacity within each interval, preserving whether the endpoint is included.

3. Distinguish a limiting rate from an attainable integer count in a finite closed observation window.

4. For minimal counts, show both a construction and why one fewer item fails the original coverage conditions.

通用步骤

1. 将依赖上取整的规则分成整数值固定的区间。

2. 在每区间内优化速率或容量，并保留端点是否包含。

3. 区分极限速率与有限闭观察窗口中可达到的整数事件数。

4. 最少数量须同时给构造与少一个物体不能满足原覆盖条件的证明。

> Example. Events are exactly fourteen minutes apart. Their phase is free. What is the greatest number occurring in a closed sixty-minute window?

> 例题：事件间隔恰为十四分钟，初相位可自由选择，六十分钟闭窗口中最多发生多少次？

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08] (continued / 续)

Worked solution

1. If q events occur, the first-to-last span is $(q-1)14$, which must be at most sixty.

2. Thus $q \le \lfloor 60/14 \rfloor + 1 = 5$.

3. Events at minutes $0, 14, 28, 42, 56$ attain five. The floor of the hourly steady rate is four, a different quantity.

例题解析

1. 若发生 q 次，首末跨度为 $(q-1)14$，必须不超过六十。

2. 所以 $q \le \lfloor 60/14 \rfloor + 1 = 5$。

3. 在零、十四、二十八、四十二、五十六分钟发生可达到五次；稳态每小时速率的整数部分为四，是另一数量。

A supremum of rates may still yield an attained integer window count; state the endpoint convention.

速率上确界即使不达到，整数窗口计数仍可能达到，须说明端点约定。

For ordered positive stamp denominations, suppose the already processed stamps have total value S and their subcollections attain every value from zero through S. Adding a stamp of value v extends coverage without a gap exactly when $v \le S+1$. Repeating this criterion gives the necessary low-denomination counts before a high denomination can be used. A final ceiling may give a construction count, but minimality requires showing a collection with one fewer stamp cannot reach the target total.

正邮票面值按序处理，若已处理邮票的总面值为 S，且其子组合能覆盖零至 S，加入面值 v 的一枚邮票能无缺口延伸，当且仅当 $v \le S+1$。反复使用即得到启用较高面值前必须具备的低面值数量。最后的上取整可给构造数量，但最小性还须证明少一枚的总面值无法达到目标。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08] (continued / 续)

Practice / 配套练习: AIME-2008-I-Q12 (WB p. 173), AIME-2022-II-Q14 (WB p. 808)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09]

Certify intersections of full folded branches

证明完整折叠分支交点计数有效

Procedure

1. Split every absolute and trigonometric fold at all extrema, zeros, and break levels; list each monotone full-range branch.

2. Use the coupled equations to confine every solution and check all branch endpoints separately.

3. Prove one crossing per branch pair: endpoint signs give existence, while orientation or a derivative-product bound gives uniqueness.

4. If a boundary corner replaces a sign change, analyze it locally and certify whether an additional interior root exists.

通用步骤

1. 在所有极值、零点与折叠高度处分开绝对值和三角分段，列全单调满值域分支。

2. 利用联立方程限制全部解，并单独检验所有分支端点。

3. 证明每对分支恰有一次穿越：端点符号给存在性，方向或导数乘积界给唯一性。

4. 若边界角点替代变号，须局部分析是否还有内部根。

> Example. On $[0,1]^2$, solve $y = x^3$ and $x = 2y$. Explain why two increasing functions need not have just one intersection.

> 例题：在单位正方形中解 $y = x^3, x = 2y$，并解释两个递增函数为何不一定只有一个交点。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

Worked solution

1. Substitute to get $x = 2x^3$, or $x(1 - 2x^2) = 0$.

2. The two admissible solutions are $(0, 0)$ and $(1/\sqrt{2}, 1/(2\sqrt{2}))$. The negative root lies outside the square.

3. Both graphs are increasing, yet a boundary root and an interior root coexist. At the interior root the derivative product is $(3x^2) \cdot 2 = 3 > 1$, giving a transverse crossing.

例题解析

1. 代入得到 $x = 2x^3$，即 $x(1 - 2x^2) = 0$。

2. 单位方形内解为 $(0, 0)$ 与 $(1/\sqrt{2}, 1/(2\sqrt{2}))$，负根在方形外。

3. 两图都递增，仍可有边界根与内部根。内部根的导数乘积为 $(3x^2) \cdot 2 = 3 > 1$，故为横截穿越。

Counting branch pairs without a uniqueness argument can give an unjustified answer.

没有唯一性证明的分支对相乘，不能保证答案成立。

Crossing lemma: for equally oriented inverse graphs, put $H(y) = G(y) - F^{-1}(y)$. At a root, $H\prime = (F\prime G\prime - 1)/F\prime$. If the derivative product exceeds one at every interior root, the sign of $H\prime$ is fixed on that branch pair. Two roots with that same nonzero crossing direction are impossible. Oppositely oriented graphs are unique directly by monotonicity.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

For the coupled trigonometric-fold example, define $h(u)=4\big||u-1/2|-1/4\big|$ on $[0,1]$ and solve $y=F(x)=h(|\sin 2\pi x|)$, $x=G(y)=h(|\cos 3\pi y|)$. The knot values of h at $0,1/4,1/2,3/4,1$ are $1,0,1,0,1$, giving sixteen full-height branches of F and twenty-four full-width branches of G; their only boundary solution is $(1,1)$. The derivative magnitude is $D=96\pi^2|\cos 2\pi x\sin 3\pi y|$. If $|\cos 2\pi x|\le 1/20$, the equations imply $y\ge .99$ and $x\ge 1-18\pi^2(.01)^2>.982$, contradicting that cosine bound. If $D\le 1$, one factor is at most $1/20$, so the sine factor must be small. The equations then force $a=1-x\le .01$, $b=1-y<.01$ and exactly $b=4\sin 2\pi a$, $a=4(1-\cos 3\pi b)$.

Let $K(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$. On $[0,.01]$, $K''=36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]>0$. Since $K(0)=K(b_*)=0$ for an interior root, strict convexity gives $K'(b_*)>0$, equivalent to $D>1$, a contradiction. Thus the crossing lemma applies. Near zero $K'(0)<0$, while $K(.01)>0$, so the corner branch also has an interior root. This supplies the needed uniqueness and corner certificate for counting the full branch pairs.

穿越引理：同向逆图令 $H(y)=G(y)-F^{-1}(y)$，零点处 $H'=(F'G'-1)/F'$。若每个内部零点导数乘积均大于一，则同一分支对上 H' 在零点具有固定非零符号，不能有两个同向穿越根。反向图则直接由单调性唯一。

联立三角折叠例中，先定义 $h(u)=4\big||u-1/2|-1/4\big|$ $(0\le u\le 1)$，求解 $y=F(x)=h(|\sin 2\pi x|)$、$x=G(y)=h(|\cos 3\pi y|)$。h 在 $0,1/4,1/2,3/4,1$ 的节点值为 $1,0,1,0,1$，故 F 有十六个满高度分支、G 有二十四个满宽度分支，唯一边界解为 $(1,1)$。导数绝对乘积为 $D=96\pi^2|\cos 2\pi x\sin 3\pi y|$。若余弦因子至多二十分之一，方程迫使 $y\ge .99$ 与 $x\ge 1-18\pi^2(.01)^2>.982$，与该余弦界矛盾。若 $D\le 1$，至少一因子不超过二十分之一，所以只能正弦因子小。方程进而迫使 $a=1-x\le .01, b=1-y<.01$，并精确满足 $b=4\sin 2\pi a, a=4(1-\cos 3\pi b)$。

令 $K(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$。在 $[0,.01]$ 上，其二阶导数 $36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]$ 为正。内部根使 $K(0)=K(b_*)=0$，严格凸性给出 $K'(b_*)>0$，恰等价于 $D>1$，产生矛盾。因此穿越引理适用。零附近 $K'(0)<0$ 而 $K(.01)>0$，故角点分支另有内部根。这补全按满分支对计数所需的唯一性与角点证书。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

Practice / 配套练习: AIME-2024-I-Q12 (WB p. 179)